

# Week 7 · Class 1: Practice Exercises — ANSWER KEY

## Multivariate Models & Categorical Predictors and Interactions

2025-12-31

### 1 Segment A — Multivariate Models

#### 2 Non-AI Exercises

##### 2.1 1. Understanding Multivariate Regression

###### 2.1.1 1.1 Multiple Choice: Why Control Variables?

Answer: b) To isolate the effect of our variable of interest

###### 2.1.2 1.2 Fill in the Blanks: Multiple Regression

1.  $\beta_1$  represents the effect of  $X_1$  holding  $\mathbf{X}$  constant
2. This is called the **partial** effect of  $X_1$
3.  $\hat{Y}$  is the predicted  $Y$  when both  $X_1$  and  $X_2$  equal **zero**
4. Adding  $X_2$  helps control for **confounding** variables
5. The model assumes  $X_1$  and  $X_2$  have **additive** effects

###### 2.1.3 1.3 True or False: Multivariate Models

**T** Control variables must be correlated with the outcome **F** Adding controls always improves causal inference **T** Multivariate models can have multiple independent variables **F** We should control for every available variable

## 2.2 2. Simpson's Paradox

### 2.2.1 2.1 Match: Simpson's Paradox Examples

a = **1** (Overall correlation reverses within subgroups = Simpson's Paradox) b = **4** (Aggregated data hides important patterns = Aggregation bias) c = **2** (A third variable explains the relationship = Confounding) d = **3** (Different groups have different relationships = Heterogeneous effects)

### 2.2.2 2.2 Code Detective: Controlling

This model controls for: `regime_type` and `gdp_per_capita`

### 2.2.3 2.3 Multiple Choice: Interpretation

Answer: b) Holding X constant, a one-unit increase in X decreases Y by 3

## 2.3 3. Model Building

### 2.3.1 3.1 Fill in the Code: Adding Controls

```
# Simple model
model1 <- lm(outcome ~ treatment, data = data)

# Add one control (age)
model2 <- lm(outcome ~ treatment + age, data = data)

# Add multiple controls (age + sex)
model3 <- lm(outcome ~ treatment + age + sex, data = data)
```

### 2.3.2 3.2 Match: Variable Selection

a = **2** (Theory-driven selection = Choose variables based on prior knowledge) b = **4** (Statistical significance = Include only "significant" predictors) c = **1** (Change in main coefficient = See if controls affect our main relationship) d = **3** (Model fit ( $R^2$ ) = Maximize explained variance)

### 2.3.3 3.3 Spot the Error: Bad Controls

**Problem:** `votes_received` is a post-treatment variable (outcome of winning). It's caused by `campaign_spending`, so controlling for it introduces post-treatment bias.

## 2.4 4. Interpreting Multivariate Results

### 2.4.1 4.1 True or False: Coefficient Changes

**F** The main coefficient always decreases **T** A large change suggests confounding **F** No change means the control is unimportant **T** The sign can flip from positive to negative **F** Standard errors always increase

### 2.4.2 4.2 Multiple Choice: Partial Effects

Answer: b) We're comparing units with the same values on control variables

### 2.4.3 4.3 Multiple Choice: $R^2$ vs. Adjusted $R^2$

Answer: b) The extra predictors add little beyond chance; adjusted  $R^2$  penalizes them, so prefer Model A

$R^2$  can never fall when predictors are added, so it cannot flag useless variables. Adjusted  $R^2$  charges a penalty per predictor and falls when additions explain less than chance would.

## 3 AI Exercises

**Tips for Working with Claude:**

- Ask for **R code using only tidyverse** (no other packages)
- Request **simple, focused answers** to your specific question—not complex analyses
- Ask Claude to **explain what the code is doing** since you're learning
- Avoid asking for visualizations or plots in these exercises
- Include the output of `glimpse()` in your prompt so Claude knows your variable names

**Example prompt:** “Using tidyverse in R, run a regression with `conflicts` as the outcome and `trade_volume` as the predictor. Keep the code simple and explain what each line does. Here is what my data looks like: [paste glimpse output]”

## 3.1 5. Trade and Conflict Analysis

### 3.1.1 5.1 Simple vs Controlled Analysis

```
trade_data <- read_csv("data/trade_conflict_data.csv")
```

```
# Simple model
```

```
simple <- lm(conflicts ~ trade_volume, data = trade_data)
```

```
tidy(simple)
```

```
# A tibble: 2 x 5
```

|   | term<br><chr> | estimate<br><dbl> | std.error<br><dbl> | statistic<br><dbl> | p.value<br><dbl> |
|---|---------------|-------------------|--------------------|--------------------|------------------|
| 1 | (Intercept)   | 0.488             | 0.0334             | 14.6               | 1.57e-40         |
| 2 | trade_volume  | -0.0000000817     | 0.0000000750       | -1.09              | 2.76e- 1         |

```
glance(simple)
```

```
# A tibble: 1 x 12
```

|   | r.squared<br><dbl> | adj.r.squared<br><dbl> | sigma<br><dbl> | statistic<br><dbl> | p.value<br><dbl> | df<br><dbl> | logLik<br><dbl> | AIC<br><dbl> | BIC<br><dbl> |
|---|--------------------|------------------------|----------------|--------------------|------------------|-------------|-----------------|--------------|--------------|
| 1 | 0.00238            | 0.000375               | 0.703          | 1.19               | 0.276            | 1           | -532.           | 1071.        | 1083.        |

```
# i 3 more variables: deviance <dbl>, df.residual <int>, nobs <int>
```

```
# With controls
```

```
controlled <- lm(conflicts ~ trade_volume + distance_km + both_democracy + alliance,  
                 data = trade_data)
```

```
tidy(controlled)
```

```
# A tibble: 5 x 5
```

|   | term<br><chr>      | estimate<br><dbl> | std.error<br><dbl> | statistic<br><dbl> | p.value<br><dbl> |
|---|--------------------|-------------------|--------------------|--------------------|------------------|
| 1 | (Intercept)        | 0.422             | 0.0734             | 5.75               | 0.0000000159     |
| 2 | trade_volume       | -0.0000000753     | 0.0000000750       | -1.00              | 0.316            |
| 3 | distance_km        | 0.00000309        | 0.00000748         | 0.413              | 0.680            |
| 4 | both_democracyTRUE | 0.142             | 0.0663             | 2.14               | 0.0330           |
| 5 | allianceTRUE       | -0.0381           | 0.0776             | -0.491             | 0.624            |

```
glance(controlled)
```

```
# A tibble: 1 x 12
  r.squared adj.r.squared sigma statistic p.value    df logLik   AIC   BIC
    <dbl>      <dbl> <dbl>      <dbl>   <dbl> <dbl> <dbl> <dbl> <dbl>
1    0.0122      0.00420 0.702        1.53   0.193     4 -530. 1072. 1097.
# i 3 more variables: deviance <dbl>, df.residual <int>, nobs <int>
```

### 3.1.2 5.2 Building Models Progressively

```
# Model 1: Just trade
m1 <- lm(conflicts ~ trade_volume, data = trade_data)

# Model 2: Add geographic/cultural controls
m2 <- lm(conflicts ~ trade_volume + distance_km + common_language + colonial_history,
         data = trade_data)

# Model 3: Add political controls
m3 <- lm(conflicts ~ trade_volume + distance_km + common_language + colonial_history +
         both_democracy + alliance, data = trade_data)

# Compare
bind_rows(
  tidy(m1) %>% filter(term == "trade_volume") %>% mutate(model = "M1: Trade only"),
  tidy(m2) %>% filter(term == "trade_volume") %>% mutate(model = "M2: + Geography"),
  tidy(m3) %>% filter(term == "trade_volume") %>% mutate(model = "M3: + Politics")
) %>% select(model, estimate, std.error, p.value)
```

```
# A tibble: 3 x 4
  model          estimate    std.error p.value
  <chr>          <dbl>      <dbl>    <dbl>
1 M1: Trade only -0.0000000817 0.0000000750 0.276
2 M2: + Geography -0.0000000803 0.0000000751 0.286
3 M3: + Politics -0.0000000718 0.0000000750 0.339
```

### 3.1.3 5.3 Testing Mechanisms

```
# Test if democracy moderates trade effect
interact_dem <- lm(conflicts ~ trade_volume * both_democracy + distance_km + alliance,
                   data = trade_data)
tidy(interact_dem)
```

```
# A tibble: 6 x 5
```

|   | term<br><chr>                   | estimate<br><dbl> | std.error<br><dbl> | statistic<br><dbl> | p.value<br><dbl> |
|---|---------------------------------|-------------------|--------------------|--------------------|------------------|
| 1 | (Intercept)                     | 0.416             | 0.0737             | 5.65               | 2.70e-8          |
| 2 | trade_volume                    | -0.0000000480     | 0.0000000814       | -0.589             | 5.56e-1          |
| 3 | both_democracyTRUE              | 0.165             | 0.0717             | 2.31               | 2.14e-2          |
| 4 | distance_km                     | 0.00000319        | 0.00000749         | 0.426              | 6.70e-1          |
| 5 | allianceTRUE                    | -0.0371           | 0.0776             | -0.478             | 6.33e-1          |
| 6 | trade_volume:both_democracyTRUE | -0.000000182      | 0.000000210        | -0.871             | 3.84e-1          |

```
# Test if alliance moderates trade effect
interact_alliance <- lm(conflicts ~ trade_volume * alliance + distance_km + both_democracy,
                        data = trade_data)
tidy(interact_alliance)
```

```
# A tibble: 6 x 5
```

|   | term<br><chr>             | estimate<br><dbl> | std.error<br><dbl> | statistic<br><dbl> | p.value<br><dbl> |
|---|---------------------------|-------------------|--------------------|--------------------|------------------|
| 1 | (Intercept)               | 0.426             | 0.0735             | 5.80               | 0.0000000120     |
| 2 | trade_volume              | -0.000000115      | 0.0000000851       | -1.35              | 0.178            |
| 3 | allianceTRUE              | -0.0666           | 0.0829             | -0.804             | 0.422            |
| 4 | distance_km               | 0.00000324        | 0.00000749         | 0.433              | 0.665            |
| 5 | both_democracyTRUE        | 0.142             | 0.0663             | 2.14               | 0.0330           |
| 6 | trade_volume:allianceTRUE | 0.000000176       | 0.000000180        | 0.980              | 0.327            |

## 3.2 6. Education and Income

### 3.2.1 6.1 Returns to Education

```
education_data <- read_csv("data/education_income_panel.csv")

# Progressive models
m1 <- lm(income ~ years_education, data = education_data)
```

```

m2 <- lm(income ~ years_education + age + experience, data = education_data)
m3 <- lm(income ~ years_education + age + experience + urban + gender + race,
         data = education_data)

# Compare education coefficients
bind_rows(
  tidy(m1) %>% filter(term == "years_education") %>% mutate(model = "M1: Education only"),
  tidy(m2) %>% filter(term == "years_education") %>% mutate(model = "M2: + Age/Experience"),
  tidy(m3) %>% filter(term == "years_education") %>% mutate(model = "M3: + All controls")
) %>% select(model, estimate, std.error, p.value)

```

```

# A tibble: 3 x 4
  model          estimate std.error p.value
  <chr>          <dbl>    <dbl>   <dbl>
1 M1: Education only    -217.     207.   0.295
2 M2: + Age/Experience  -218.     208.   0.294
3 M3: + All controls    -227.     212.   0.284

```

### 3.2.2 6.2 Omitted Variable Bias

```

# Without ability
no_ability <- lm(income ~ years_education + age + experience, data = education_data)

# With ability
with_ability <- lm(income ~ years_education + age + experience + ability_score,
                  data = education_data)

# With ability and family background
full <- lm(income ~ years_education + age + experience + ability_score + parent_education,
          data = education_data)

# Compare
bind_rows(
  tidy(no_ability) %>% filter(term == "years_education") %>% mutate(model = "No ability"),
  tidy(with_ability) %>% filter(term == "years_education") %>% mutate(model = "+ Ability"),
  tidy(full) %>% filter(term == "years_education") %>% mutate(model = "+ Family background")
) %>% select(model, estimate, std.error)

```

```

# A tibble: 3 x 3

```

|   | model               | estimate | std.error |
|---|---------------------|----------|-----------|
|   | <chr>               | <dbl>    | <dbl>     |
| 1 | No ability          | -218.    | 208.      |
| 2 | + Ability           | -216.    | 208.      |
| 3 | + Family background | -204.    | 211.      |

### 3.2.3 6.3 Heterogeneous Effects

```
# By gender
gender_int <- lm(income ~ years_education * gender + age + experience,
                 data = education_data)
tidy(gender_int) %>% filter(grepl("years_education", term))
```

```
# A tibble: 2 x 5
```

|   | term                       | estimate | std.error | statistic | p.value |
|---|----------------------------|----------|-----------|-----------|---------|
|   | <chr>                      | <dbl>    | <dbl>     | <dbl>     | <dbl>   |
| 1 | years_education            | -272.    | 290.      | -0.936    | 0.349   |
| 2 | years_education:genderMale | 115.     | 416.      | 0.275     | 0.783   |

```
# By race
race_int <- lm(income ~ years_education * race + age + experience,
               data = education_data)
tidy(race_int) %>% filter(grepl("years_education", term))
```

```
# A tibble: 5 x 5
```

|   | term                         | estimate | std.error | statistic | p.value |
|---|------------------------------|----------|-----------|-----------|---------|
|   | <chr>                        | <dbl>    | <dbl>     | <dbl>     | <dbl>   |
| 1 | years_education              | 300.     | 540.      | 0.556     | 0.579   |
| 2 | years_education:raceBlack    | -827.    | 729.      | -1.13     | 0.257   |
| 3 | years_education:raceHispanic | -684.    | 703.      | -0.973    | 0.331   |
| 4 | years_education:raceOther    | -510.    | 741.      | -0.688    | 0.491   |
| 5 | years_education:raceWhite    | -349.    | 679.      | -0.514    | 0.608   |

```
# By urban/rural
urban_int <- lm(income ~ years_education * urban + age + experience,
                data = education_data)
tidy(urban_int) %>% filter(grepl("years_education", term))
```

```
# A tibble: 2 x 5
```

|   | term                      | estimate | std.error | statistic | p.value |
|---|---------------------------|----------|-----------|-----------|---------|
|   | <chr>                     | <dbl>    | <dbl>     | <dbl>     | <dbl>   |
| 1 | years_education           | -395.    | 303.      | -1.30     | 0.192   |
| 2 | years_education:urbanTRUE | 268.     | 423.      | 0.633     | 0.527   |

### 3.3 7. Media Effects with Controls

#### 3.3.1 7.1 Experimental Analysis with Covariates

```
media_data <- read_csv("data/media_experiment_full.csv")

# Without covariates
simple_ate <- lm(attitude_change ~ treatment, data = media_data)
tidy(simple_ate)
```

```
# A tibble: 3 x 5
```

|   | term              | estimate | std.error | statistic | p.value |
|---|-------------------|----------|-----------|-----------|---------|
|   | <chr>             | <dbl>    | <dbl>     | <dbl>     | <dbl>   |
| 1 | (Intercept)       | 0.0168   | 0.163     | 0.103     | 0.918   |
| 2 | treatmentNegative | -0.0147  | 0.221     | -0.0663   | 0.947   |
| 3 | treatmentPositive | 0.0252   | 0.227     | 0.111     | 0.912   |

```
# With covariates
covariate_ate <- lm(attitude_change ~ treatment + pretreatment_attitude +
  political_knowledge + age + political_interest,
  data = media_data)
tidy(covariate_ate) %>% filter(term == "treatmentTRUE")
```

```
# A tibble: 0 x 5
# i 5 variables: term <chr>, estimate <dbl>, std.error <dbl>, statistic <dbl>,
# p.value <dbl>
```

```
# Compare precision
cat("Simple model SE:", tidy(simple_ate)$std.error[2], "\n")
```

Simple model SE: 0.2210866

```
cat("Covariate model SE:", tidy(covariate_ate)$std.error[2], "\n")
```

Covariate model SE: 0.2226879

### 3.3.2 7.2 Improving Precision

```
# Test heterogeneous effects by party strength
het_party <- lm(attitude_change ~ treatment * party_strength +
               pretreatment_attitude + political_knowledge + age,
               data = media_data)
tidy(het_party) %>% filter(grepl("treatment", term))
```

```
# A tibble: 5 x 5
```

|   | term<br><chr>                    | estimate<br><dbl> | std.error<br><dbl> | statistic<br><dbl> | p.value<br><dbl> |
|---|----------------------------------|-------------------|--------------------|--------------------|------------------|
| 1 | treatmentNegative                | -0.134            | 0.523              | -0.256             | 0.798            |
| 2 | treatmentPositive                | -0.590            | 0.547              | -1.08              | 0.281            |
| 3 | pretreatment_attitude            | -0.0174           | 0.0515             | -0.338             | 0.735            |
| 4 | treatmentNegative:party_strength | 0.0291            | 0.116              | 0.251              | 0.802            |
| 5 | treatmentPositive:party_strength | 0.150             | 0.118              | 1.27               | 0.206            |

```
# By education
het_edu <- lm(attitude_change ~ treatment * education +
              pretreatment_attitude + political_knowledge + age,
              data = media_data)
tidy(het_edu) %>% filter(grepl("treatment", term))
```

```
# A tibble: 9 x 5
```

|   | term<br><chr>                           | estimate<br><dbl> | std.error<br><dbl> | statistic<br><dbl> | p.value<br><dbl> |
|---|-----------------------------------------|-------------------|--------------------|--------------------|------------------|
| 1 | treatmentNegative                       | 0.309             | 0.435              | 0.709              | 0.478            |
| 2 | treatmentPositive                       | 0.635             | 0.424              | 1.50               | 0.135            |
| 3 | pretreatment_attitude                   | -0.0132           | 0.0517             | -0.255             | 0.799            |
| 4 | treatmentNegative:educationGraduate     | -0.796            | 0.606              | -1.31              | 0.190            |
| 5 | treatmentPositive:educationGraduate     | -1.45             | 0.617              | -2.35              | 0.0189           |
| 6 | treatmentNegative:educationHigh School  | 0.0485            | 0.633              | 0.0766             | 0.939            |
| 7 | treatmentPositive:educationHigh School  | -0.155            | 0.630              | -0.246             | 0.806            |
| 8 | treatmentNegative:educationSome College | -0.353            | 0.640              | -0.552             | 0.581            |
| 9 | treatmentPositive:educationSome College | -0.806            | 0.652              | -1.24              | 0.217            |

```
# By age
het_age <- lm(attitude_change ~ treatment * age +
              pretreatment_attitude + political_knowledge + party_strength,
              data = media_data)
tidy(het_age) %>% filter(grepl("treatment", term))
```

```
# A tibble: 5 x 5
  term                estimate std.error statistic p.value
  <chr>                <dbl>     <dbl>     <dbl>   <dbl>
1 treatmentNegative   -0.0162    0.649    -0.0249  0.980
2 treatmentPositive    0.0898    0.676     0.133   0.894
3 pretreatment_attitude -0.0134    0.0516   -0.260   0.795
4 treatmentNegative:age  0.000282   0.0131    0.0216  0.983
5 treatmentPositive:age -0.00114   0.0138   -0.0826  0.934
```

```
# Check covariate balance
media_data %>%
  group_by(treatment) %>%
  summarize(
    mean_pretreat = mean(pretreatment_attitude),
    mean_knowledge = mean(political_knowledge),
    mean_party = mean(party_strength),
    mean_age = mean(age),
    mean_interest = mean(political_interest)
  )
```

```
# A tibble: 3 x 6
  treatment mean_pretreat mean_knowledge mean_party mean_age mean_interest
  <chr>      <dbl>         <dbl>     <dbl>   <dbl>         <dbl>
1 Control    3.74          5.18      4.21    45.9          5.26
2 Negative   4.21          5.45      3.91    47.7          5.49
3 Positive   4.01          4.55      4.19    46.6          5.85
```

### 3.3.3 7.3 Specification Choices

```
# Different specifications
spec1 <- lm(attitude_change ~ treatment, data = media_data)
spec2 <- lm(attitude_change ~ treatment + pretreatment_attitude, data = media_data)
spec3 <- lm(attitude_change ~ treatment + pretreatment_attitude + political_knowledge,
```

```

      data = media_data)
spec4 <- lm(attitude_change ~ treatment + pretreatment_attitude + political_knowledge +
  party_strength + age, data = media_data)
spec5 <- lm(attitude_change ~ treatment + pretreatment_attitude + political_knowledge +
  party_strength + age + education + news_consumption + political_interest,
  data = media_data)

# Compare treatment effects
bind_rows(
  tidy(spec1) %>% filter(grepl("treatment", term)) %>% mutate(spec = "No controls"),
  tidy(spec2) %>% filter(grepl("treatment", term)) %>% mutate(spec = "+ Pretreatment"),
  tidy(spec3) %>% filter(grepl("treatment", term)) %>% mutate(spec = "+ Knowledge"),
  tidy(spec4) %>% filter(grepl("treatment", term)) %>% mutate(spec = "+ Party/Age"),
  tidy(spec5) %>% filter(grepl("treatment", term)) %>% mutate(spec = "Full model")
) %>% select(spec, estimate, std.error, p.value)

```

```

# A tibble: 14 x 4
  spec      estimate std.error p.value
  <chr>      <dbl>    <dbl>  <dbl>
1 No controls -0.0147    0.221   0.947
2 No controls  0.0252    0.227   0.912
3 + Pretreatment -0.00809  0.223   0.971
4 + Pretreatment  0.0290    0.228   0.899
5 + Pretreatment -0.0140    0.0513  0.784
6 + Knowledge   -0.0110    0.223   0.961
7 + Knowledge    0.0356    0.229   0.876
8 + Knowledge   -0.0140    0.0513  0.786
9 + Party/Age   -0.00225   0.224   0.992
10 + Party/Age  0.0371    0.229   0.871
11 + Party/Age  -0.0135    0.0514  0.793
12 Full model   0.0435    0.223   0.846
13 Full model   0.100     0.229   0.662
14 Full model  -0.0247    0.0516  0.633

```

## 3.4 8. Campaign Finance Multivariate

### 3.4.1 8.1 Money in Politics

```

campaign_data <- read_csv("data/campaign_complete.csv")

```

```
# Comprehensive model
full_model <- lm(vote_share ~ total_spending + incumbent + challenger_quality +
                partisan_lean + president_approval + unemployment + primary_contested,
                data = campaign_data)
tidy(full_model)
```

```
# A tibble: 9 x 5
  term                estimate std.error statistic  p.value
  <chr>                <dbl>    <dbl>    <dbl>    <dbl>
1 (Intercept)         53.5        3.40      15.8 1.44e-45
2 total_spending     -0.000000299 0.000000204 -1.46 1.44e- 1
3 incumbentTRUE        1.10        1.06       1.04 2.98e- 1
4 challenger_qualityLow  0.103        1.23       0.0840 9.33e- 1
5 challenger_qualityMedium -1.52        1.29      -1.18 2.37e- 1
6 partisan_lean       -0.00964      0.0329     -0.293 7.70e- 1
7 president_approval  -0.0217      0.0595     -0.365 7.15e- 1
8 unemployment        -0.399      0.256     -1.56 1.20e- 1
9 primary_contestedTRUE -0.0402      1.17     -0.0344 9.73e- 1
```

```
glance(full_model)
```

```
# A tibble: 1 x 12
  r.squared adj.r.squared sigma statistic p.value    df logLik   AIC   BIC
  <dbl>      <dbl> <dbl>    <dbl>    <dbl> <dbl> <dbl> <dbl> <dbl>
1  0.0166    0.000608 11.6      1.04    0.406     8 -1928. 3877. 3919.
# i 3 more variables: deviance <dbl>, df.residual <int>, nobs <int>
```

### 3.4.2 8.2 Relative Importance

```
# Standardize continuous variables
campaign_std <- campaign_data %>%
  mutate(
    total_spending_z = scale(total_spending)[,1],
    partisan_lean_z = scale(partisan_lean)[,1],
    president_approval_z = scale(president_approval)[,1],
    unemployment_z = scale(unemployment)[,1]
  )

# Model with standardized variables
```

```
std_model <- lm(vote_share ~ total_spending_z + partisan_lean_z + president_approval_z +
               unemployment_z + incumbent + challenger_quality + primary_contested,
               data = campaign_std)
tidy(std_model)
```

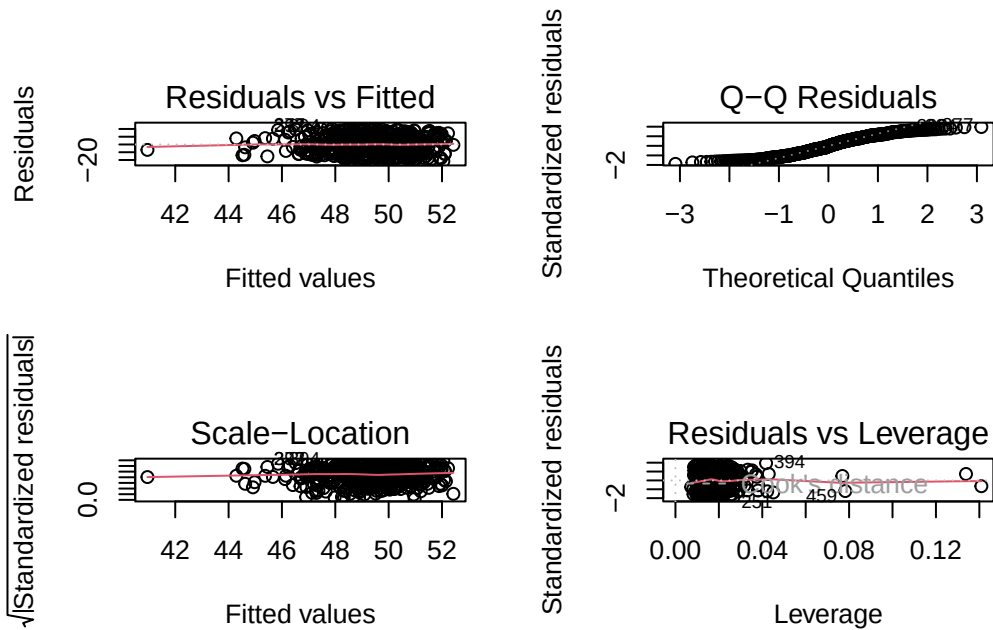
```
# A tibble: 9 x 5
  term                estimate std.error statistic  p.value
  <chr>                <dbl>    <dbl>    <dbl>    <dbl>
1 (Intercept)         49.3      0.982     50.2 1.84e-195
2 total_spending_z    -0.764    0.521    -1.46 1.44e- 1
3 partisan_lean_z     -0.153    0.521    -0.293 7.70e- 1
4 president_approval_z -0.191    0.524    -0.365 7.15e- 1
5 unemployment_z     -0.812    0.521    -1.56 1.20e- 1
6 incumbentTRUE        1.10     1.06     1.04 2.98e- 1
7 challenger_qualityLow 0.103     1.23     0.0840 9.33e- 1
8 challenger_qualityMedium -1.52     1.29    -1.18 2.37e- 1
9 primary_contestedTRUE -0.0402    1.17    -0.0344 9.73e- 1
```

```
# Test interactions
interact_model <- lm(vote_share ~ total_spending * incumbent +
                    partisan_lean * president_approval + challenger_quality +
                    unemployment + primary_contested,
                    data = campaign_data)
tidy(interact_model) %>% filter(grepl(":", term))
```

```
# A tibble: 2 x 5
  term                estimate  std.error statistic p.value
  <chr>                <dbl>    <dbl>    <dbl>    <dbl>
1 total_spending:incumbentTRUE -0.0000000707 0.000000466   -0.152 0.880
2 partisan_lean:president_approval -0.00370    0.00380   -0.975 0.330
```

### 3.4.3 8.3 Diagnostic Checks

```
# Residual plots
par(mfrow = c(2, 2))
plot(full_model)
```



```
# Influential observations
campaign_data %>%
  mutate(
    cooks_d = cooks.distance(full_model),
    leverage = hatvalues(full_model)
  ) %>%
  arrange(desc(cooks_d)) %>%
  select(candidate_id, vote_share, total_spending, incumbent, cooks_d, leverage) %>%
  head(10)
```

# A tibble: 10 x 6

|    | candidate_id | vote_share | total_spending | incumbent | cooks_d | leverage |
|----|--------------|------------|----------------|-----------|---------|----------|
|    | <dbl>        | <dbl>      | <dbl>          | <lgl>     | <dbl>   | <dbl>    |
| 1  | 394          | 69.7       | 11003672.      | FALSE     | 0.0183  | 0.0417   |
| 2  | 459          | 31.3       | 16043714.      | FALSE     | 0.0136  | 0.0782   |
| 3  | 251          | 30.6       | 2933350.       | TRUE      | 0.0108  | 0.0267   |
| 4  | 371          | 30.1       | 12129645.      | FALSE     | 0.00968 | 0.0451   |
| 5  | 341          | 52.1       | 21990735.      | FALSE     | 0.00906 | 0.134    |
| 6  | 236          | 65.9       | 447662.        | FALSE     | 0.00793 | 0.0242   |
| 7  | 457          | 34.0       | 22823683.      | FALSE     | 0.00774 | 0.141    |
| 8  | 376          | 31.4       | 1192920.       | TRUE      | 0.00761 | 0.0252   |
| 9  | 410          | 30.6       | 11679385.      | FALSE     | 0.00739 | 0.0423   |
| 10 | 254          | 31.6       | 1553790.       | FALSE     | 0.00737 | 0.0261   |

## 4 Segment B — Categorical Predictors and Interactions

### 5 Non-AI Exercises

#### 5.1 1. Categorical Variables in Regression

##### 5.1.1 1.1 Multiple Choice: Dummy Variables

Answer: c) 3

(For k categories, we create k-1 dummy variables)

##### 5.1.2 1.2 Fill in the Blanks: Reference Categories

1. The excluded category is called the **reference** category
2. Coefficients represent differences from the **reference category**
3. With k categories, we include **k-1** dummy variables
4. The choice of reference category affects **interpretation**, not **model fit**
5. R automatically creates dummies for **factor** variables

##### 5.1.3 1.3 True or False: Categorical Predictors

**F** We must convert text categories to numbers before regression **T** The reference category has a coefficient of zero **T** Factor levels should be meaningful, not arbitrary **T** We can include all k dummies if we remove the intercept

#### 5.2 2. Interpreting Categorical Coefficients

##### 5.2.1 2.1 Code Detective: Factor Regression

The 5.2 means: **Engineering employees earn \$5.2k more than the reference department (likely the first alphabetically, which would be a department not shown, or the intercept represents the base department)**

##### 5.2.2 2.2 Match: Interpretation

a = **2** (Intercept = 50 means reference group's average is 50) b = **4** (genderFemale = -5 means females score 5 points lower than males) c = **1** (partyRepublican = 10 means Republicans score 10 points higher than Democrats) d = **3** (educationGrad = 15 means graduates score 15 points higher than reference)

### 5.2.3 2.3 Spot the Error: Dummy Trap

**Problem:** Perfect multicollinearity. If someone is not male, they must be female (male + female = 1 always). We can only include one dummy (e.g., just “female”) or use both without an intercept.

## 5.3 3. Interaction Terms

### 5.3.1 3.1 Multiple Choice: Interactions

Answer: b) The effect of X depends on X

### 5.3.2 3.2 Fill in the Code: Creating Interactions

```
# Main effects only
model1 <- lm(y ~ x1 + x2, data = data)

# With interaction (x2)
model2 <- lm(y ~ x1 * x2, data = data) # Shorthand
```

### 5.3.3 3.3 True or False: Interaction Effects

**T**  $X * X$  automatically includes main effects in R **F** Interactions must be between continuous variables **T** The main effect of X is its effect when  $X = 0$  **T** Centering variables changes interaction interpretation **F** Significant interactions mean main effects don't matter

## 5.4 4. Interpreting Interactions

### 5.4.1 4.1 Code Detective: Interaction Meaning

Effect =  $2 + (-0.5)(4) = 2 - 2 = 0$

(The marginal effect of X =  $2 + X = 2 - 0.5(4) = 0$ )

### 5.4.2 4.2 Match: Interaction Types

a = **2** (Continuous  $\times$  Continuous = Age and experience interact for salary) b = **1** (Continuous  $\times$  Categorical = Income effect varies by gender) c = **3** (Categorical  $\times$  Categorical = Party and region interact for voting)

### 5.4.3 4.3 Multiple Choice: Marginal Effects

Answer: b) Depends on the value of X

## 6 AI Exercises

### Tips for Working with Claude:

- Ask for **R code using only tidyverse** (no other packages)
- Request **simple, focused answers** to your specific question—not complex analyses
- Ask Claude to **explain what the code is doing** since you’re learning
- Avoid asking for visualizations or plots in these exercises
- Include the output of `glimpse()` in your prompt so Claude knows your variable names

**Example prompt:** “Using tidyverse in R, run a regression predicting political\_engagement from party. Keep the code simple and explain what each line does. Here is what my data looks like: [paste glimpse output]”

### 6.1 5. Party and Demographics

#### 6.1.1 5.1 Categorical Predictors

```
party_data <- read_csv("data/party_demographics.csv")

# Model political engagement with categorical predictors
m1 <- lm(political_engagement ~ party, data = party_data)
tidy(m1)
```

```
# A tibble: 4 x 5
  term          estimate std.error statistic  p.value
<chr>          <dbl>    <dbl>    <dbl>    <dbl>
1 (Intercept)    50.3      1.63     30.8 2.29e-117
2 partyIndependent -0.305    2.42    -0.126 9.00e- 1
3 partyOther     -1.41    3.83    -0.369 7.12e- 1
4 partyRepublican  1.25    2.20     0.569 5.69e- 1
```

```
m2 <- lm(political_engagement ~ party + age_group + education, data = party_data)
tidy(m2)
```

```
# A tibble: 10 x 5
```

|    | term<br><chr>         | estimate<br><dbl> | std.error<br><dbl> | statistic<br><dbl> | p.value<br><dbl> |
|----|-----------------------|-------------------|--------------------|--------------------|------------------|
| 1  | (Intercept)           | 51.5              | 2.92               | 17.6               | 3.07e-54         |
| 2  | partyIndependent      | -0.504            | 2.43               | -0.207             | 8.36e- 1         |
| 3  | partyOther            | -2.00             | 3.86               | -0.519             | 6.04e- 1         |
| 4  | partyRepublican       | 0.789             | 2.21               | 0.357              | 7.21e- 1         |
| 5  | age_group30-44        | -4.21             | 2.63               | -1.60              | 1.11e- 1         |
| 6  | age_group45-64        | -1.69             | 2.54               | -0.668             | 5.05e- 1         |
| 7  | age_group65+          | -4.88             | 2.69               | -1.82              | 6.95e- 2         |
| 8  | educationGraduate     | 4.30              | 2.56               | 1.68               | 9.31e- 2         |
| 9  | educationHS           | 1.84              | 2.63               | 0.699              | 4.85e- 1         |
| 10 | educationSome College | 0.169             | 2.70               | 0.0626             | 9.50e- 1         |

### 6.1.2 5.2 Reference Category Effects

```
# Default reference (alphabetically first)
m_default <- lm(political_engagement ~ age_group, data = party_data)
tidy(m_default)
```

```
# A tibble: 4 x 5
```

|   | term<br><chr>  | estimate<br><dbl> | std.error<br><dbl> | statistic<br><dbl> | p.value<br><dbl> |
|---|----------------|-------------------|--------------------|--------------------|------------------|
| 1 | (Intercept)    | 53.2              | 1.91               | 27.9               | 1.60e-103        |
| 2 | age_group30-44 | -4.26             | 2.62               | -1.62              | 1.05e- 1         |
| 3 | age_group45-64 | -1.59             | 2.53               | -0.627             | 5.31e- 1         |
| 4 | age_group65+   | -4.52             | 2.66               | -1.70              | 8.98e- 2         |

```
# Change reference to 65+
party_data_relevel <- party_data %>%
  mutate(age_group = factor(age_group, levels = c("65+", "18-29", "30-44", "45-64")))

m_relevel <- lm(political_engagement ~ age_group, data = party_data_relevel)
tidy(m_relevel)
```

```
# A tibble: 4 x 5
```

|   | term<br><chr>  | estimate<br><dbl> | std.error<br><dbl> | statistic<br><dbl> | p.value<br><dbl> |
|---|----------------|-------------------|--------------------|--------------------|------------------|
| 1 | (Intercept)    | 48.7              | 1.85               | 26.3               | 4.96e-96         |
| 2 | age_group18-29 | 4.52              | 2.66               | 1.70               | 8.98e- 2         |

|   |                |       |      |       |          |
|---|----------------|-------|------|-------|----------|
| 3 | age_group30-44 | 0.263 | 2.58 | 0.102 | 9.19e- 1 |
| 4 | age_group45-64 | 2.93  | 2.49 | 1.18  | 2.39e- 1 |

```
# Interpretation: Coefficients now show differences relative to 65+
```

## 6.2 6. Gender and Career Outcomes

### 6.2.1 6.1 Gender Main Effects

```
career_data <- read_csv("data/career_gender_interact.csv")

# Gender salary gap with controls
gender_model <- lm(salary ~ gender + years_experience + department +
  performance_rating + education_level,
  data = career_data)
tidy(gender_model) %>% filter(grepl("gender", term))
```

```
# A tibble: 2 x 5
  term          estimate std.error statistic p.value
<chr>          <dbl>    <dbl>    <dbl>   <dbl>
1 genderMale      0.284      2.03     0.140   0.889
2 genderNon-binary 1.75       4.62     0.378   0.705
```

### 6.2.2 6.2 Testing for Interactions

```
# Test if experience returns differ by gender
exp_gender <- lm(salary ~ years_experience * gender + department +
  performance_rating + education_level,
  data = career_data)
tidy(exp_gender) %>% filter(grepl("years_experience|gender", term))
```

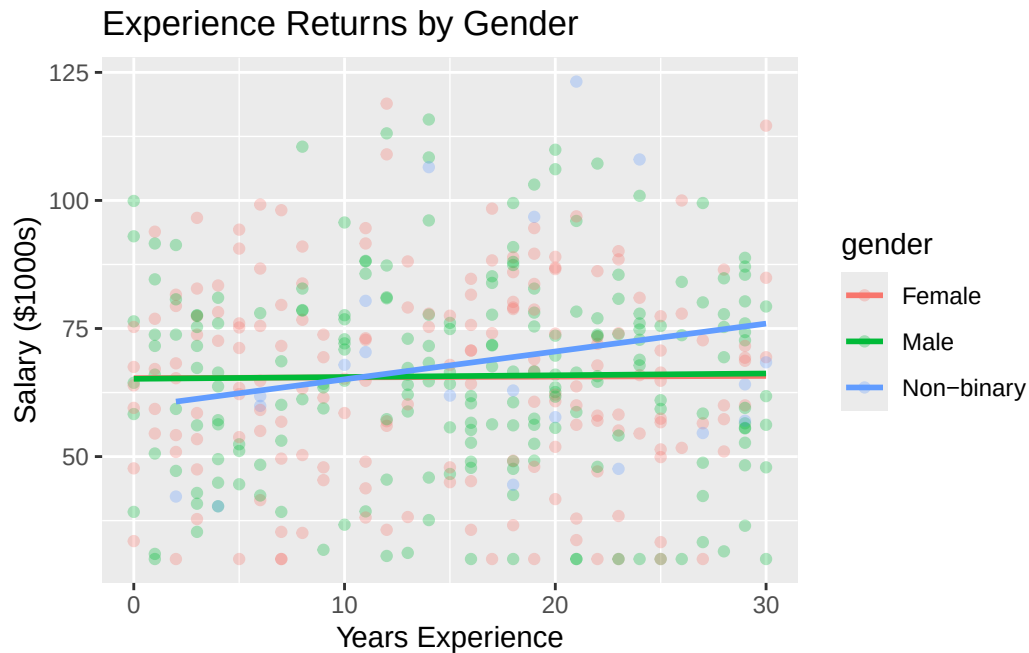
```
# A tibble: 5 x 5
  term          estimate std.error statistic p.value
<chr>          <dbl>    <dbl>    <dbl>   <dbl>
1 years_experience  0.0321    0.170     0.189   0.851
2 genderMale      0.553     4.01     0.138   0.890
3 genderNon-binary -6.81    10.2     -0.669   0.504
4 years_experience:genderMale -0.0167  0.231    -0.0722  0.942
5 years_experience:genderNon-binary 0.509    0.545     0.935   0.350
```

```
# Test department × gender interaction
dept_gender <- lm(salary ~ department * gender + years_experience +
                  performance_rating + education_level,
                  data = career_data)
tidy(dept_gender) %>% filter(grepl("department.*gender|gender.*department", term))
```

```
# A tibble: 8 x 5
```

|   | term<br><chr>                        | estimate<br><dbl> | std.error<br><dbl> | statistic<br><dbl> | p.value<br><dbl> |
|---|--------------------------------------|-------------------|--------------------|--------------------|------------------|
| 1 | departmentFinance:genderMale         | -6.88             | 6.92               | -0.995             | 0.320            |
| 2 | departmentHR:genderMale              | -1.64             | 6.88               | -0.239             | 0.811            |
| 3 | departmentMarketing:genderMale       | -0.957            | 6.80               | -0.141             | 0.888            |
| 4 | departmentSales:genderMale           | 4.53              | 6.62               | 0.684              | 0.494            |
| 5 | departmentFinance:genderNon-binary   | 32.5              | 20.0               | 1.62               | 0.105            |
| 6 | departmentHR:genderNon-binary        | 5.15              | 16.4               | 0.314              | 0.754            |
| 7 | departmentMarketing:genderNon-binary | -11.2             | 17.5               | -0.642             | 0.521            |
| 8 | departmentSales:genderNon-binary     | 0.946             | 17.0               | 0.0556             | 0.956            |

```
# Visualize experience × gender interaction
career_data %>%
  ggplot(aes(x = years_experience, y = salary, color = gender)) +
  geom_point(alpha = 0.3) +
  geom_smooth(method = "lm", se = FALSE) +
  labs(title = "Experience Returns by Gender",
        x = "Years Experience", y = "Salary ($1000s)")
```



## 6.3 7. Regional Political Differences

### 6.3.1 7.1 Regional Categories

```
regional_data <- read_csv("data/regional_politics.csv")

# Model vote share by region
region_model <- lm(dem_vote_share ~ region, data = regional_data)
tidy(region_model)
```

```
# A tibble: 4 x 5
  term          estimate std.error statistic  p.value
<chr>          <dbl>    <dbl>    <dbl>    <dbl>
1 (Intercept)    46.4      3.03     15.3 1.05e-19
2 regionNortheast  3.50      4.38      0.800 4.28e- 1
3 regionSouth     5.77      4.13      1.40 1.69e- 1
4 regionWest      2.58      4.20      0.614 5.42e- 1
```

```
# With demographic controls
full_model <- lm(dem_vote_share ~ region + college_percent + urban_percent +
                 evangelical_percent + union_membership,
                 data = regional_data)
tidy(full_model)
```

```
# A tibble: 8 x 5
  term                estimate std.error statistic    p.value
  <chr>                <dbl>    <dbl>    <dbl>    <dbl>
1 (Intercept)         57.6      10.6      5.44 0.00000251
2 regionNortheast      2.01      4.72     0.426 0.672
3 regionSouth          5.94      4.36     1.36 0.180
4 regionWest           2.81      4.76     0.591 0.557
5 college_percent     -0.220     0.207    -1.06 0.294
6 urban_percent       -0.00322    0.0838   -0.0384 0.970
7 evangelical_percent -0.0826     0.131    -0.631 0.532
8 union_membership    -0.161     0.371    -0.434 0.667
```

### 6.3.2 7.2 Region × Demographics

```
# Test if college effect varies by region
college_region <- lm(dem_vote_share ~ college_percent * region + urban_percent,
                    data = regional_data)
tidy(college_region) %>% filter(grepl("college|region", term))
```

```
# A tibble: 7 x 5
  term                estimate std.error statistic p.value
  <chr>                <dbl>    <dbl>    <dbl>    <dbl>
1 college_percent     -0.138     0.387   -0.358    0.722
2 regionNortheast      2.44     18.3     0.133    0.895
3 regionSouth         11.4     21.0     0.543    0.590
4 regionWest          10.3     18.3     0.561    0.578
5 college_percent:regionNortheast 0.00131    0.581    0.00226    0.998
6 college_percent:regionSouth  -0.177     0.613   -0.289    0.774
7 college_percent:regionWest  -0.272     0.553   -0.493    0.625
```

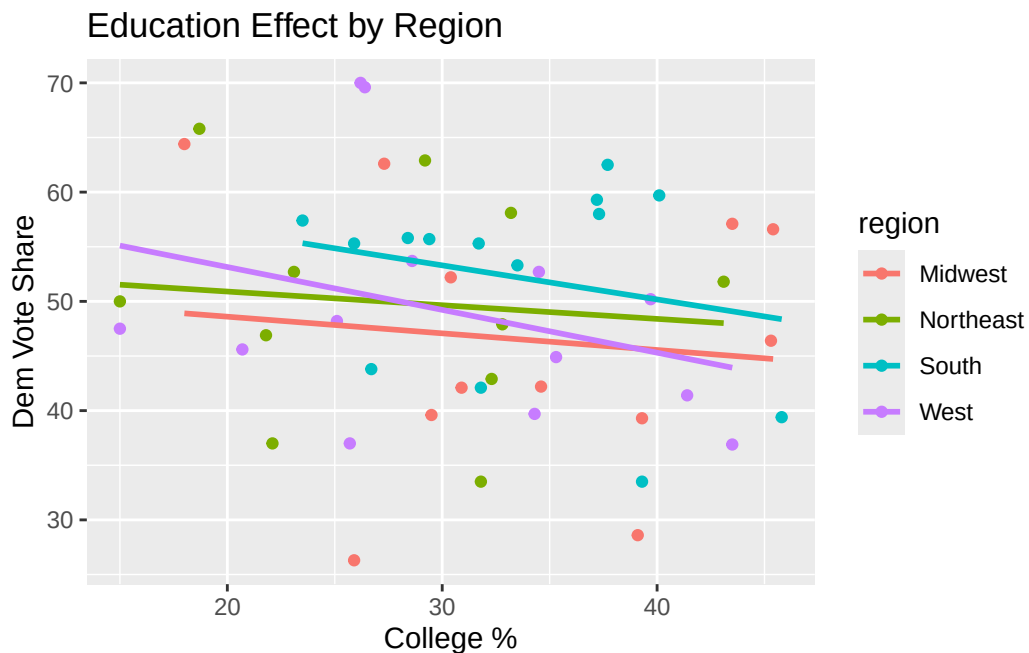
```
# Test if urban effect varies by region
urban_region <- lm(dem_vote_share ~ urban_percent * region + college_percent,
                  data = regional_data)
tidy(urban_region) %>% filter(grepl("urban|region", term))
```

```
# A tibble: 7 x 5
```

|   | term                          | estimate | std.error | statistic | p.value |
|---|-------------------------------|----------|-----------|-----------|---------|
|   | <chr>                         | <dbl>    | <dbl>     | <dbl>     | <dbl>   |
| 1 | urban_percent                 | -0.0200  | 0.182     | -0.109    | 0.913   |
| 2 | regionNortheast               | -5.52    | 17.5      | -0.316    | 0.754   |
| 3 | regionSouth                   | 1.84     | 17.6      | 0.104     | 0.917   |
| 4 | regionWest                    | 13.7     | 19.2      | 0.712     | 0.481   |
| 5 | urban_percent:regionNortheast | 0.103    | 0.238     | 0.433     | 0.667   |
| 6 | urban_percent:regionSouth     | 0.0547   | 0.243     | 0.225     | 0.823   |
| 7 | urban_percent:regionWest      | -0.182   | 0.267     | -0.682    | 0.499   |

```
# Visualize
```

```
regional_data %>%  
  ggplot(aes(x = college_percent, y = dem_vote_share, color = region)) +  
  geom_point() +  
  geom_smooth(method = "lm", se = FALSE) +  
  labs(title = "Education Effect by Region",  
       x = "College %", y = "Dem Vote Share")
```



## 6.4 8. Policy Implementation Effects

### 6.4.1 8.1 Implementation Categories

```
policy_data <- read_csv("data/policy_heterogeneity.csv")

# Model outcome by implementation quality
quality_model <- lm(outcome ~ implementation_quality, data = policy_data)
tidy(quality_model)
```

```
# A tibble: 3 x 5
```

|   | term                         | estimate | std.error | statistic | p.value   |
|---|------------------------------|----------|-----------|-----------|-----------|
|   | <chr>                        | <dbl>    | <dbl>     | <dbl>     | <dbl>     |
| 1 | (Intercept)                  | 49.4     | 1.47      | 33.7      | 2.16e-103 |
| 2 | implementation_qualityLow    | 0.449    | 2.12      | 0.212     | 8.32e- 1  |
| 3 | implementation_qualityMedium | -1.28    | 2.11      | -0.605    | 5.45e- 1  |

```
# With controls
controlled_model <- lm(outcome ~ implementation_quality + state_capacity +
  political_support + prior_policies,
  data = policy_data)
tidy(controlled_model)
```

```
# A tibble: 6 x 5
```

|   | term                         | estimate | std.error | statistic | p.value  |
|---|------------------------------|----------|-----------|-----------|----------|
|   | <chr>                        | <dbl>    | <dbl>     | <dbl>     | <dbl>    |
| 1 | (Intercept)                  | 45.3     | 3.85      | 11.7      | 2.28e-26 |
| 2 | implementation_qualityLow    | 0.0976   | 2.14      | 0.0455    | 9.64e- 1 |
| 3 | implementation_qualityMedium | -1.50    | 2.14      | -0.703    | 4.82e- 1 |
| 4 | state_capacity               | 1.65     | 4.36      | 0.379     | 7.05e- 1 |
| 5 | political_support            | 0.0460   | 0.0453    | 1.02      | 3.11e- 1 |
| 6 | prior_policies               | 0.398    | 0.502     | 0.793     | 4.29e- 1 |

### 6.4.2 8.2 Context-Dependent Effects

```
# Test urban/rural differences in policy effects
urban_policy <- lm(outcome ~ policy_implemented * urban_rural + state_capacity +
  political_support,
```

```

      data = policy_data)
tidy(urban_policy) %>% filter(grepl("policy|urban", term))

```

```

# A tibble: 5 x 5
  term                                estimate std.error statistic p.value
  <chr>                                <dbl>    <dbl>    <dbl>   <dbl>
1 policy_implementedTRUE              1.31      2.90      0.450   0.653
2 urban_ruralSuburban                -3.50      3.06     -1.14   0.254
3 urban_ruralUrban                    2.20      3.14      0.698   0.486
4 policy_implementedTRUE:urban_ruralSuburb~ 2.76      4.25      0.649   0.517
5 policy_implementedTRUE:urban_ruralUrban  -3.63      4.25     -0.855   0.393

```

```

# Test quality × support interaction
quality_support <- lm(outcome ~ implementation_quality * political_support +
  state_capacity + urban_rural,
  data = policy_data)
tidy(quality_support) %>% filter(grepl("quality|support", term))

```

```

# A tibble: 5 x 5
  term                                estimate std.error statistic p.value
  <chr>                                <dbl>    <dbl>    <dbl>   <dbl>
1 implementation_qualityLow          -3.92e-1    6.45     -0.0608  0.952
2 implementation_qualityMedium       -1.15e+1    6.26     -1.84    0.0667
3 political_support                  -9.91e-3    0.0737    -0.135   0.893
4 implementation_qualityLow:political_supp~ 1.24e-2    0.110      0.113   0.910
5 implementation_qualityMedium:political_s~ 1.84e-1    0.108      1.69    0.0918

```

```

# Visualize policy effects by context
policy_data %>%
  ggplot(aes(x = political_support, y = outcome, color = implementation_quality)) +
  geom_point(alpha = 0.3) +
  geom_smooth(method = "lm", se = FALSE) +
  facet_wrap(~ urban_rural) +
  labs(title = "Quality × Support Interaction by Urban/Rural",
       x = "Political Support", y = "Outcome")

```

Quality × Support Interaction by Urban/Rural

